# ASHWIN SATISH

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## EDUCATION

**Northeastern University - Khoury College of Computer Sciences** **Sept 2024 - Dec 2026 | Boston, MA**
Master of Science, Data Science **GPA**: 3.83/4.0
*Machine Learning, Natural Language Processing, MLOps, Algorithms, Data Mining, Database Management, Capstone*

**Christ University** **June 2020 - May 2023 | Bangalore, India**
Bachelor of Science, Computer Science **GPA**: 3.53/4.0

## EXPERIENCE

**Institute for Experiential AI, Northeastern University |** Machine Learning Developer Intern **Jan 2026 - Present | Boston, MA**

- Engineered an **LLM** pipeline, mining **35M** records, finding **278K**+ publication-trial links & transforming **13K**+ papers into structured data via **prompt engineering** & strict **JSON Schema** outputs, with every field traced to its source sentence.
- Evaluated & benchmarked **LLMs** on **AWS Bedrock** for clinical-data extraction across cost, recall, & token efficiency, cutting per-paper inference cost **~65%** while retaining **90% recall** against the baseline model.
- Built a fallback LLM pipeline on **AWS Bedrock** with automated error handling, adaptive retries & batch inference recovery, achieving **95%** recovery across failed records.
- Architected a custom entity-normalization pipeline mapping **45K+** clinical-trial adverse events to medical ontologies, raising coverage from **75.1%** to **85.4%** (**+4.6K** events) with zero regressions.
- Designed & deployed a clinical-trial data model in Neo4j (**115K**+ nodes, **116K**+ relationships), enabling structured retrieval of trial data through REST APIs.

**Northeastern University |** Machine Learning Assistant **Jan 2025 - Dec 2025 | Boston, MA**

- Boosted true positives from **2% to 77%** (maintaining zero false positives**)** for Lyme cell detection via 7-model CNN ensemble.
- Built a **Python** image preprocessing pipeline for **30,000+** MATLAB cell images, saving the biology team weeks of manual work.
- Improved ensemble accuracy by **13%** via augmentation & L2/dropout regularization, validated with k-fold cross-validation.
- Mentored & conducted office hours as a Teaching Assistant for **300+ students** in Machine Learning, statistics & **NumPy** demos.

**CarerAI |** Data Science Intern **Jul 2023 - May 2024 | Gurgaon, India**

- Prevented **2,000+ unnecessary procedures** for customers by uncovering **500+** surgery avoidance cases through k-Nearest Neighbors (KNN) analysis of patient records against historical data.
- Leveraged EHR for a risk score, improving classification accuracy by **26.4%**, increasing confidence in surgery avoidance cases.
- Improved KNN's score by **20%** using DBSCAN to remove record outliers, reducing false positives in surgery recommendations.

## SKILLS

**Languages**: Python, SQL, R, Java, TypeScript, JavaScript, Cypher, HTML, CSS
**Libraries & Frameworks**: Scikit-learn, TensorFlow, PyTorch, Flask, Django, OpenCV, Pandas, NumPy, pytest, Hugging Face
**Cloud & Data**: AWS, Amazon Bedrock, Google Cloud Platform (GCP), Azure, PostgreSQL, Docker, Neo4j, Git
**ML & LLMs**: Agentic AI, LoRA, Model Context Protocol (MCP), LangChain, LangGraph, Ollama, PydanticAI, MLflow

## PROJECTS

**Multi-Agent Pokemon Battle Strategy Engine |** *Ollama, Streamlit, ChromaDB, BM25, Dense Retrieval, RAG* | **GitHub**

- Built a hybrid **Multi-Agent System** (**MAS**) where a deterministic engine computes every fact & a verifier rejects contradicting **LLM** answers, raising Llama 3B's accuracy from **26% to 97.5%** on a blind held out benchmark.
- Engineered a **RAG** pipeline with hybrid **BM25** + dense retrieval, contextual chunk headers & format scoped filtering over **2.2K Smogon** passages, raising top 1 retrieval accuracy from **74% to 94%**.

**Pantry & Recipe Assistant |** ***2nd Place - Google AI Expo*** **|** *LLM Ops, PostgreSQL, Airflow, Docker, GCP, Vertex AI* **| GitHub**

- Built an LLM & data pipeline with VLM receipt OCR for constraint-aware recipes, with automated evals, bias slicing, **CI** smoke tests, & Slack-triggered retraining on **Google Cloud Run**.
- Fine-tuned Llama 3.2 3B (**LoRA** + **DPO**, Llama 3 70B as teacher model for labels) on **12K** recipes, cutting diet violations **50%**.

**F1 Race Strategy & Tyre Degradation Simulator |** *Monte Carlo, LightGBM, Bayesian, Scikit-learn, Pandas* **| GitHub**

- Engineered a 20 car **Monte Carlo** race engine that simulates every lap, pit stop, safety car & overtake, predicting finishing order **48**% more accurately than grid position & recommending the best pit strategy based on circuit & tyre compound.
- Modeled tyre wear on **500+** sessions (LightGBM & Bayes), cutting lap error **17%** & **adapting** race by race for 2026's new cars.